Patrick Marmaroli

Freelance Speech & Audio ML Engineer
Signal processing to production software

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PROFESSIONAL SUMMARY

Signal processing engineer with a PhD from EPFL, 15+ years of experience, 25+ publications (7 journal articles, 740+ citations) and 2 granted US patents. I build speech and audio AI that works outside the lab, from signal processing and model fine-tuning to the production software around it, for demanding domains where errors are costly. An AI-assisted development workflow lets me own a project end to end with the same engineering rigor. Freelancing full-time since August 2026 through Hidacs Sàrl, fully remote from the Geneva area; co-founder of Vocametrix (speech analysis platform for speech-language pathologists).

WORK EXPERIENCE

Co-Founder — Feb 2016 — Present
Hidacs Sàrl · Switzerland · Remote — side project; full-time Oct–Dec 2021 and since Aug 2026

- Freelance missions in speech & audio ML, signal processing and production software: feasibility studies, proofs of concept, model improvement and end-to-end builds.
- Business platform suite (ERP, e-signature, timesheets, single sign-on) for a multi-company railway infrastructure alliance.
- Phoneme recognition models for clinical speech therapy (Wav2Vec2, LoRA / DoRA fine-tuning).

Co-Founder — Oct 2023 — Present
Vocametrix OÜ · Tallinn, Estonia

- Designed and deployed a speech-therapy web platform combining pronunciation assessment, pitch tracking, and gamified voice exercises, backed by a full speech-processing API suite serving both the platform and external developers.

Senior Audio Engineer — Jan 2022 — Jul 2026
Microsoft (Skype/Teams) · Tallinn, Estonia

- Developed PyTorch-based models to evaluate audio signal quality across diverse acoustic environments.
- Designed automated tools for large-scale audio and video data analysis, generating comprehensive evaluation reports.
- Conducted acoustic laboratory measurements and experimental validation to support research on audio conferencing quality.

Audio Signal Processing Consultant — Oct 2021 — Dec 2021
Onfido

- Developed an anti-spoofing technology using ultrasound for identity-verification liveness detection.

Research Engineer (part-time) — Jul 2020 — Sep 2021
HEPIA (HES-SO Geneva) · Geneva, Switzerland

- Contributed to the "EARSENS" research project, developing signal processing algorithms to repurpose loudspeakers as safety sensors for privacy-preserving indoor activity detection.

Audio Department Manager May 2016 — Aug 2021
HED Technologies · Geneva, Switzerland

- Contributed to the acoustic design and testing of audio devices, focusing on noise reduction and echo cancellation.
- Developed custom acoustic test benches and technical specifications, ensuring high-fidelity performance in complex sound environments.
- Built and led a team of 4 engineers from the ground up, steering signal processing algorithms from initial prototype to validated product.

Project Manager, R&D Acoustics Department Sep 2014 — Jun 2016
PRONA SA · Yverdon, Switzerland

- Developed custom Matlab pipelines to streamline RT60, STC, and NRC data processing, increasing the efficiency of building and environmental acoustic audits.
- Conducted on-site normative acoustic measurements and authored technical reports ensuring compliance with international standards.
- Managed multidisciplinary acoustic consulting projects for industrial, institutional, and private clients.

Post-Doctoral Fellow Mar 2013 — Oct 2014
EPFL · Lausanne, Switzerland

- Designed audio signal processing algorithms and conducted laboratory and field experiments within European research projects.
- Collaborated with industrial partners including Phonak (hearing aids), FIAT, and ARMASUISSE.
- Supervised interns and PhD students, contributed to scientific publications, and coordinated with international teams.

SELECTED PATENTS & PUBLICATIONS

US Patent 10,149,074 — Hearing assistance system (2018)

US Patent 9,510,090 — Device and method for capturing and processing voice (2016)

Virtual speech therapist: a clinician-in-the-loop AI speech therapy agent for personalized and supervised therapy — arXiv:2605.01101 (2026)

Towards the detection and classification of indoor events using a loudspeaker — Applied Acoustics (2023)

EDUCATION

PhD, Acoustics & Audio Signal Processing 2008 — 2013
EPFL · Lausanne, Switzerland

Thesis: Bimodal Sound Source Tracking Applied to Road Traffic Monitoring

M.Sc. in Signal Processing 2007 — 2008
Université du Sud Toulon-Var · France

Signal processing, psychoacoustics, and trajectography

Engineering Degree (Diplôme d'ingénieur), Software 2005 — 2007
ISEN Toulon · France

SKILLS

Core: Speech & Audio ML · Signal Processing & Acoustic Measurement · Production Software · AI-Assisted Development

Audio & DSP: Speech Recognition, Phoneme Recognition, POLQA / MOS, MATLAB, GCC-PHAT, HRTF

AI & ML: PyTorch, Wav2Vec2, LoRA / DoRA Fine-Tuning, MLflow, DVC, Scikit-learn, YAMNet, LLM APIs (Claude, GPT), Azure Cognitive Services

Web & Cloud: React, Next.js, Node.js/Express, Azure, PostgreSQL, Docker, CI/CD

Languages: French (native), English (C1 Advanced)